

Sample End Semester Question Paper - 2025

Subject: Operating Systems | Code: CS401 | Semester: 4

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Name the four Coffman conditions.
2. What is paging?
3. Name two page replacement algorithms.
4. Define thrashing.
5. What is a PCB?
6. Name four functions of an operating system.
7. What is a process?
8. State the basic states of a process.
9. Define turnaround time.
10. What is a deadlock?

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain file systems: directory structures, allocation methods and free-space management.
2. Explain system calls and the user-kernel interface.
3. Explain process states and transitions with a diagram. Explain the PCB.
4. Explain CPU scheduling: FCFS, SJF, Priority and Round Robin. Compute average waiting time for a given set of processes.
5. Explain deadlocks: conditions, prevention, avoidance and detection with examples.
6. Solve the Banker's algorithm for a given allocation and maximum matrix.
7. Explain memory management: contiguous allocation, paging and segmentation.
8. Explain page replacement: FIFO, OPT and LRU. Compare them with an example.
9. Explain virtual memory and demand paging. Discuss thrashing.
10. Explain inter-process communication: pipes, message passing and shared memory.